Lesson 4: Project: Spreadsheets with AI

Module 1 | 60 minutes | Data, formulas, checks | Adults (18+) · ages 18–65

Objective: Use AI to understand and clean a sales sheet, write formulas like SUMIF, XLOOKUP and IF, build a summary and chart, and verify every total yourself.

Success criteria

- I can ask AI to explain a sheet and list data problems.
- I can get and use formulas like SUMIF, IF and XLOOKUP from a plain-words request.
- I can make a summary table and a chart.
- I can verify AI totals with my own formulas and explain any difference.

Key vocabulary

- **CSV file**: A simple spreadsheet file (comma-separated values) that Excel, Google Sheets and AI tools can open.
- **formula**: An instruction in a cell that calculates something, starting with =.
- **SUMIF**: Adds up only the rows that match a condition, like one region.
- **XLOOKUP / VLOOKUP**: Finds a value in another table, like a product's price.
- **duplicate**: The same row entered twice, which makes totals too big.
- **verify**: Check a number yourself before you trust or share it.

You will need

- Projector or screen share with Kitabash and the AI tools for this lesson
- Devices: 1 per learner (laptop, tablet or phone)
- Learner accounts only where allowed: see the course safety notes
- Printed worksheet (optional)
- Printed project brief
- Project files: monthly_sales.csv, l04_spreadsheet_project_kit.md, l04_teacher_demo_and_answer_key.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game**	Human spreadsheet with a duplicate card (see Unplugged).
4 min	**Watch + poll**	Play the project video. Poll: "How confident are you with Excel or Google Sheets? Show 1–5 fingers."
7 min	**Live demo**	Upload, list problems, the wrong total, clean, SUMIF for West, summary and chart.
5 min	**Game: needs cleaning?**	Needs cleaning or fine? sort, then the formula match, played as a quick team race.

6 min	**Prompt Lab**	Prompt Lab: ask for a formula. Learners paste it into the AI and type the formula into their sheet.
18 min	**Project build (pairs)**	Build-along steps: understand, clean, formulas, summary, chart. Halfway check: everyone has a cleaned sheet and one working SUMIF.
5 min	**Verify and compare**	“Which total do you trust?” then pairs compare their checked totals with another pair; any difference must be explained.
6 min	**Show-and-tell**	Three learners show their chart and one insight they checked.
4 min	**Probe and submit**	New situation: real customer data. Submit the module project with the reflection.

Unplugged warm-up

Human spreadsheet (5 min): six volunteers hold cards with a city, region and amount (one duplicate card, one with “bangalore”). The class adds the South total out loud. Then reveal the duplicate and recount. Ask: “Which total would you send to your manager?”

Teaching the idea

Spreadsheets scare many adults, and AI removes the fear: you describe what you want and it writes the formula and explains it. But spreadsheets are also where AI mistakes are most dangerous, because a wrong total looks exactly like a right one. Teach the order: understand the sheet, clean it, write formulas, summarise and chart, and verify every total with your own SUM. Use the practice file only: it has a duplicate invoice, a city spelt two ways and a blank row, so the AI’s quick total is wrong in a way learners can catch.

Analogy: AI with a spreadsheet is like a fast junior accountant: brilliant at formulas and tables, but you still tally the final total before it goes to the boss.

Live demo: 1) Open monthly_sales.csv in Google Sheets or Excel on screen. 2) Upload it to Claude or ChatGPT with: “Explain each column in simple words, then list any data problems.” 3) Show the three problems in the sheet. 4) Ask ChatGPT, before cleaning, “What is the total revenue?” and type =SUM(J2:J42) in the sheet; compare. 5) Delete the duplicate INV-2431 and show the new SUM (₹5,23,971). 6) Ask Gemini or any chatbot for the SUMIF for West and type it (₹81,935). 7) Build a region summary and Insert → Chart. 8) If Gemini in Sheets or Copilot in Excel is available on your account, show it briefly; otherwise say “If you don’t see it, a chatbot tab works just as well.”

Questions to ask

Ask	Listen for
What should you do before adding up any totals?	Clean the data: duplicates, spelling differences and blanks.
What does SUMIF do, in plain words?	Adds up only the rows that match a condition.

The AI and your SUM disagree. Where do you look first?	Duplicates, blanks and spelling differences.
How do you get a formula you do not know?	Describe the columns and what you want, and ask the AI to explain it simply.
Can you upload your company's real sales file?	Not to a personal account. Use practice or anonymised data, or your company's approved tool.

Differentiation

Support
Give a half-built summary table with the region names already typed; learners ask AI for the formula and fill the totals. Pair learners new to spreadsheets with confident ones.

Stretch
Ask the AI for a pivot table guide and build revenue by region × category; add an IF column that flags sales reps below a target you choose.

Common misconceptions

Misconception	How to fix it
Computers do not make maths mistakes, so AI totals are right.	AI can miscount or include messy rows. Verify with your own SUM.
I need to memorise formulas.	Describe what you want; ask the AI to write and explain the formula.
Cleaning data is optional.	Duplicates and blanks change totals. Clean first, count second.
If Copilot or Gemini is not in my sheet, I cannot use AI.	Use any chatbot in another tab and paste the formula into your sheet.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Data cleaning	Totals made on uncleaned data	One or two problems fixed	All three problems fixed and noted	All fixed, noted, and the effect on totals explained
Formulas	No formulas, numbers typed by hand	One formula works	Three formulas work and are explained	Formulas work, are explained, and are reused for other regions or categories

Summary and chart	No summary	Summary or chart, but unclear	Clear summary table and titled chart	Clear, well-labelled, and chosen to answer the manager’s question
Verification and honesty	AI numbers used without checking	Some totals checked	All key totals checked and any difference explained	Checked, explained, and AI use clearly labelled with the time saved noted

Exit ticket
The AI’s total and your SUM differ by exactly one row’s revenue. What is the most likely reason?

Homework
Finish the module project. Optional: take a spreadsheet from your own work, remove all names and personal details, and ask the AI for one useful formula.

Answer key — Kitabash activities

1. Prompt Lab: ask for a formula (perform · ai lab)

Checklist: Name your app · Describe the columns · Say what you want calculated · Say the condition (which rows) · Ask for a simple explanation

Strong example:

```
In Google Sheets, Category is in column G and Revenue is in column J. Give me a formula
to add up the total revenue for the Home category only, and explain it in simple words
so I can change it for other categories.
```

2. Look at monthly_sales.csv. Does this need cleaning before you add up revenue? (perform · categorize)

Needs cleaning: INV-2431 appears twice, “bangalore” is spelt differently from “Bengaluru”, INV-2426 has no units and no revenue
Fine as it is: Prices are written without the ₹ sign, Dates are written like 2026-09-01, Several sales happened on the same day

3. Match each formula to the job it does (perform · matching)

SUMIF → Add up revenue only for the South region · XLOOKUP or VLOOKUP → Find a product’s price from another table · IF → Show “Target met” when a sale is above ₹20,000 · COUNTIF → Count how many invoices came from Pune

4. Which total do you trust? (perform · multiple choice)

Answer: Find the difference (₹18,720), see it is the duplicate invoice, and use your checked total

5. Module project: analyse the monthly sales sheet (probe · project)

6. New situation: real customer data (probe · multiple choice)

Answer: Remove names and phone numbers (or replace them with codes) and share only the columns needed, like product and amount

Answer key — worksheet

1. **Key words:** CSV file = A simple spreadsheet file (comma-separated values) that Excel, Google Sheets and AI tools can open. · formula = An instruction in a cell that calculates something, starting with =. · SUMIF = Adds up only the rows that match a condition, like one region. · XLOOKUP / VLOOKUP = Finds a value in another table, like a product’s price.

2. **Prompt Lab: ask for a formula:** Include: Name your app; Describe the columns; Say what you want calculated; Say the condition (which rows); Ask for a simple explanation. Example: In Google Sheets, Category is in column G and Revenue is in column J. Give me a formula to add up the total revenue for the Home category only, and explain it in simple words so I can change it for other categories.
3. **Look at monthly_sales.csv. Does this need cleaning before you add up revenue?:** Needs cleaning: INV-2431 appears twice, “bangalore” is spelt differently from “Bengaluru”, INV-2426 has no units and no revenue · Fine as it is: Prices are written without the ₹ sign, Dates are written like 2026-09-01, Several sales happened on the same day
4. **Match each formula to the job it does:** SUMIF → Add up revenue only for the South region · XLOOKUP or VLOOKUP → Find a product’s price from another table · IF → Show “Target met” when a sale is above ₹20,000 · COUNTIF → Count how many invoices came from Pune
5. **Which total do you trust?:** B. Find the difference (₹18,720), see it is the duplicate invoice, and use your checked total — The difference is exactly one duplicate row. Your checked total on clean data is the one you report, and you can explain why.
6. **My plan:** A clear plan that covers every requirement.
7. **Checklist:** All requirements met.
8. **Two stars and a wish:** Kind, specific feedback and an improvement.

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